Yousef Alaa Awad

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Engineering with University Honors | GPA: 3.77 — Expected Aug 2027
Relevant Courses: Verification of Digital Systems, HDL in Digital Systems, and Adv. Computer Architecture
Affiliations: IEEE-HKN (Zeta Chi Chapter), IEEE, ACM, Burnett Honors College

Work Experience

Advanced Micro Devices, Inc. — Austin, TX
Validation/Debug Intern — Sept 2026 – Present

- Developing a Python issue-triage pipeline integrating Jira REST APIs and the Anthropic Claude API to sync debug results and context across 2 validation teams, eliminating manual copy-paste and saving 5–10 minutes per triaged ticket.
- Reimaging 15–20 post-silicon SoC validation systems multiple times weekly via an automated flashing suite, deploying matched BIOS, firmware, OS, and driver versions so regression failures isolate to genuine test failures or external factors.
- Developing a Python script for the reimaging suite that collapses a multi-step manual stage of the flow into a single automated step, reducing per-cycle operator intervention across the validation farm.

Lockheed Martin Corporation — Orlando, FL
ASIC/FPGA Design CWEP Intern — Feb 2026 – July 2026

- Built a Formal Property Verification (FPV) flow via Synopsys VC Formal from scratch, authoring 20+ SVAs for bounded model checking, catching 3 critical bugs (illegal state transitions, deadlocks) missed by existing functional tests.
- Extended the flow with Formal Security Verification (FSV/SPV) by authoring security properties for an encryption module, formally verifying the module against possible secret leakage.
- Deployed the FPV/FSV suite in GitLab CI/CD with a modular, config-driven structure, cutting per-module formal flow setup from 30–60 minutes to ∼5 minutes via a parameterized testbench template and a single pipeline variable toggle.

Embedded Software Engineering CWEP Intern — Apr 2025 – Feb 2026

- Designed a system-level Hardware-in-the-Loop (HIL) simulation environment in C++, enabling high-fidelity testing within the existing workflow for 20+ software engineers by implementing a CIGI v2 compliant image generator.
- Remediated 250+ security violations flagged by static code analysis, resolving critical memory-safety issues including buffer overflows and unsafe pointer access across the codebase.
- Engineered the migration of legacy C++/Qt tooling environments from Visual Studio to a CMake build system, slashing compilation times by 50% and establishing a modular architecture to improve future scalability of the application.

Projects

KnightCore – RISC-V SoC – 2025 AMD Hardware Competition – Group Project — Apr 2025 – Aug 2025

- Developed a coverage-driven verification testbench in Python (Cocotb) to validate a custom 32-bit RISC-V SoC (RV32I ISA), implementing constrained-random stimulus to effectively explore the design state space.
- Developed a comprehensive test suite with directed tests for all ISA-defined instructions and constrained-random stimulus generation to stress the ALU and branch prediction logic, leading to the discovery and resolution of 5 critical RTL bugs.

2025 SouthEastCon Robotics Competition – Embedded Software Lead – Group Project — Sep 2024 – Apr 2025

- Managed Git/GitHub workflows with branching strategies, code reviews, and issue tracking to ensure team-wide consistency. Built CI/CD pipelines for automated testing and deployment, and implemented HIL testing for early hardware validation.
- Led the embedded software pipeline using ROS2, enabling low-latency control between modular hardware and decision nodes, which was then validated via Gazebo/OpenCV, contributing to 1st place in Design and 2nd place in Performance.

CyndaQuil Compiler/Language – Solo Project — Sep 2024 – Jan 2025

- Designed and implemented a statically-typed compiled language from scratch in C++, building the full toolchain (parser, semantic analyzer, and x86 codegen) with CMake managing a multi-stage build pipeline.
- Engineered low-level system components and optimized performance, integrating Assembly and C to enhance hardware-software interfacing, ensuring minimal latency and efficient computation for embedded systems.

Leadership

Eta Kappa Nu (IEEE-HKN)
Chapter President, Zeta Chi Chapter — Apr 2026 – Present

- Directing chapter operations and budget following Key Chapter recognition, expanding induction classes, driving alumni re-engagement, and scaling member programming to ensure long-term chapter sustainability.

Technical Skills

Verification & Formal Methods: Synopsys VC Formal, SystemVerilog Assertions (SVA), Bounded Model Checking, Formal Property Verification (FPV), Formal Security Verification (FSV/SPV), UVM, Constrained-Random Verification
Languages: SystemVerilog, Verilog, C++, Tcl, Python, C, Bash, Rust, Assembly (x86, RISC-V, MIPS)
Computer Architecture: RISC-V ISA, AXI/Memory-Mapped Interfaces, Pipelining, Caching, Memory Hierarchy
Hardware & EDA Tools: Synopsys VCS, Xilinx Vivado, Verilator
Systems & Scripting: Linux, Git, GitLab CI/CD, ROS/ROS2, Real-Time C++, CMake/Make